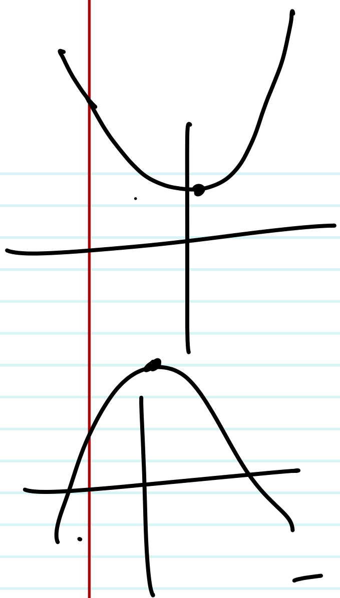
Math.1



$x^2 - 12x + 37 = 0$

$\Rightarrow x^2 - 2 \times 6 \times x + 36 + 1 = 0$

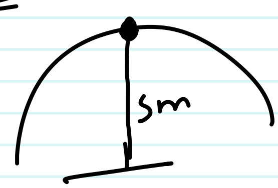
$\Rightarrow (x-6)^2 + 1 = 0$

$\Rightarrow (x-6)^2 = -1$ $\quad \sqrt{-1}$

$-8x = -2 \times 4 \times x +$

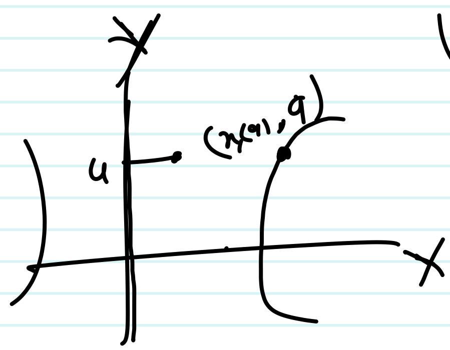
$\boxed{h(t) = -5t^2 + \frac{1}{2}ut}$

$t = -\frac{b}{2a}$



$\boxed{S = 5 \times \left(\frac{-b}{2a}\right) + \frac{1}{2}u\left(\frac{-b}{2a}\right)}$

$y = x^2 + \cdots$



$y = a$

$x(a)$

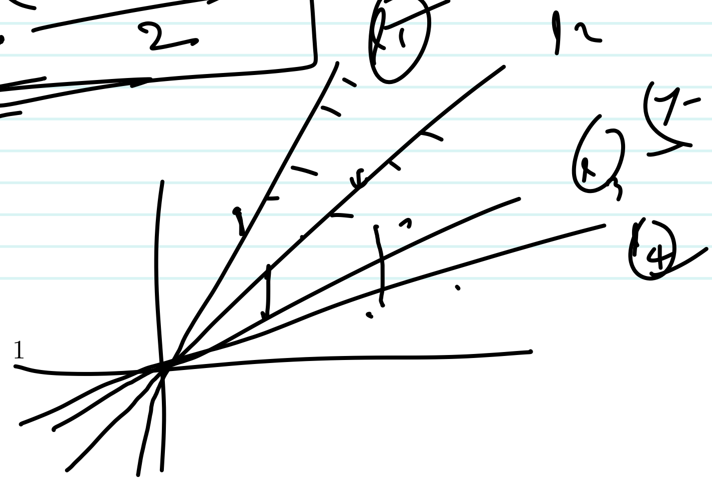
$h(t) = at^2 + bt + c$

$h(t+1) = a(t+1)^2 + b(t+1) + c$

$= at^2 + 2at + a + bt + b + c$

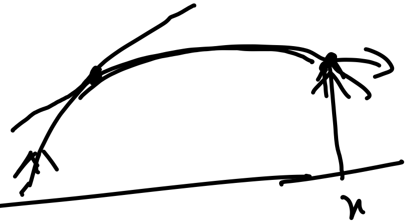
$t+1, \quad t$

$\boxed{h(t+1) - h(t) = 4 \cdot \frac{(t+1)+t}{2}}$



$f(x) = ax^2+bx+c$

$\boxed{2ax+b}$



slope question

$f'(x)$

$-\frac{2}{125}x = -0.64$

$a=-3$

$f(x) = ax^2+bx+c$

$a>0$

$y = a(x+2)(b-x)$

$= a(bx - x^2 + 2b - 2x)$

$y = -ax^2 + (b-2)ax + 2ab$

$-a>0$

$a<0$

$-a>0 \Rightarrow \boxed{a<0}$

$\boxed{-a>0}$

$\boxed{-a>0} \Rightarrow \boxed{a<0}$

(-1)

$a>b$

$\boxed{-a<-b}$

Deb $\qquad$ ANC

$\alpha + (20-\alpha) = 20$

$20-\alpha-4 = 16-\alpha$

$\alpha-3$

Deb

$20 - \boxed{3} = \boxed{x = \alpha-3}$

$20 - \boxed{4}$ ANC $= y = 16-\alpha$

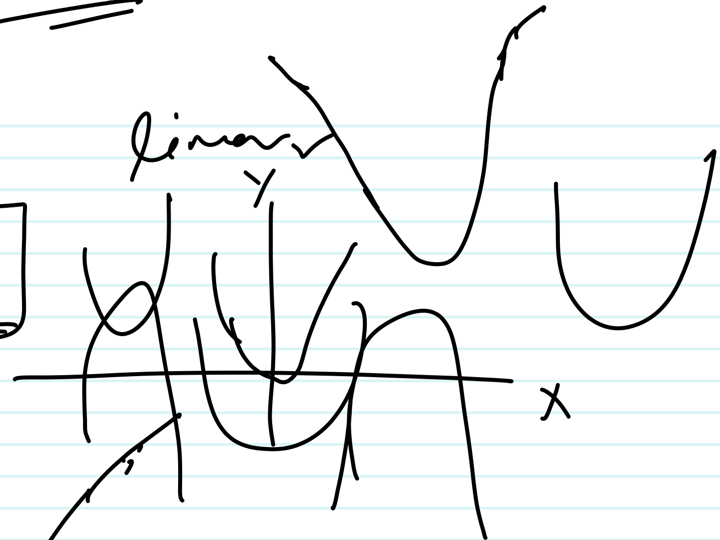
$xy = 42$

$(\alpha-3)(16-\alpha) = 42$

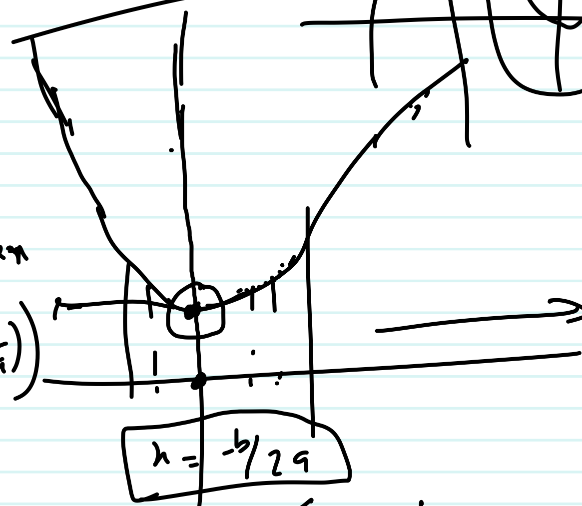
$\Rightarrow$ variable

Summary

$f(x) = ax+b$ linear



$\boxed{f(x) = ax^2+bx+c}$



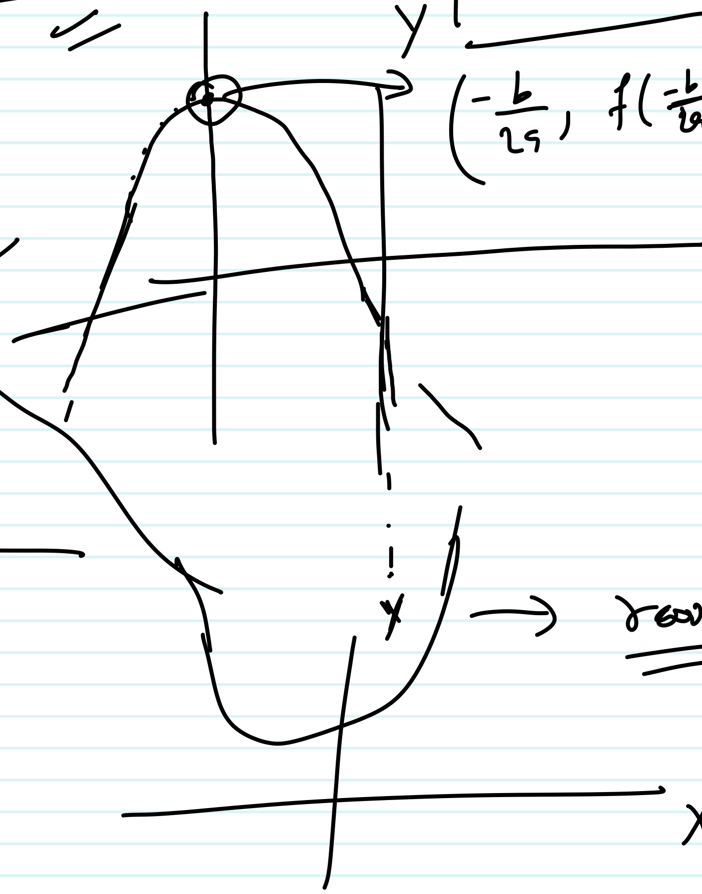
vertex

$\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$

$\boxed{x = -b/2a}$

$f(x) = ax^2+bx+c$

$\boxed{x = -\frac{b}{2a}}$

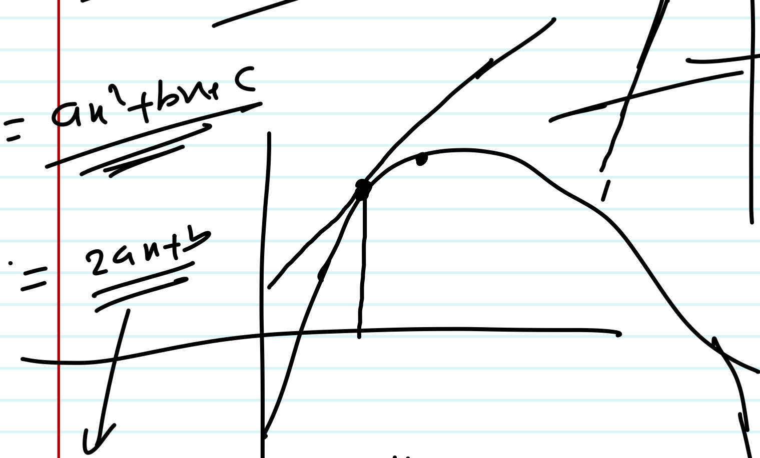


$\left(-\frac{b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

slope of Quad

$f(x) = ax^2+bx+c$

slope $f(x) = 2ax+b$



3

$2a\times3+b = 6a+b$

roots → X intercept

Y intercept

foil method

$f(x) = x^2+6x-16 =$

$x^2+8x-2x-16$

8×2

$8-2=6$

3

$(x-3)$ factor

3 and poly $(x-3)$ factor

Square method

parabola

$a^2+2ab+b^2$

$(a+b)^2$

$2\times 3\times x$

$g(x) = x^2 + 6x - 16$

$= x^2 + 6x + 9 - 9 - 16$

$= (x+3)^2 - 25 = 0$

$(x+3)^2 = 25$

$\Rightarrow x+3 = \pm\sqrt{25}$

$x+3 = \pm 5$

$\Rightarrow x+3 = 5$ or $x+3=-5$

$x = 2$ $\quad x = -8$

$(x+3)^2 - 5^2 = 0$

$\Rightarrow (x+3+5)(x+3-5) = 0$

$\Rightarrow (x+8)(x-2) = 0$

$= x(x+8) - 2(x+8)$

$= (x+8)(x-2) = 0$

$x+8=0$ $\quad x-2=0$

$x=-8$ or $x=2$

a^2-b^2

$=(a+b)(a-b)$

$(a+b)^2 = a^2+2ab+b^2$

$x^2+6x-16$

$2\times x\times 3$

$x^2+2\times x\times 3 - 16 + 9 - 9$

$x^2+2\times x\times 3+9-25$

$x^2+2\times x\times 3+3^2 - 25$

$(x+3)^2 - 25$

$f(x) = ax^2+bx+c = 0$

$\Rightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

$\Rightarrow x^2 + \frac{2}{2}\frac{b}{a}x + \frac{c}{a} =$

$(a+b)^2 = a^2+2ab+b^2$

$x^2 + 2\times\left(\frac{b}{2a}\right)\times x + \frac{c}{a} + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 = 0$

$\Rightarrow x^2 + 2\times\frac{b}{2a}\times x + \left(\frac{b}{2a}\right)^2 + \frac{c}{a} - \left(\frac{b}{2a}\right)^2 = 0$

$$\Rightarrow \left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \left(\frac{b}{2a}\right)^2 = 0$$

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

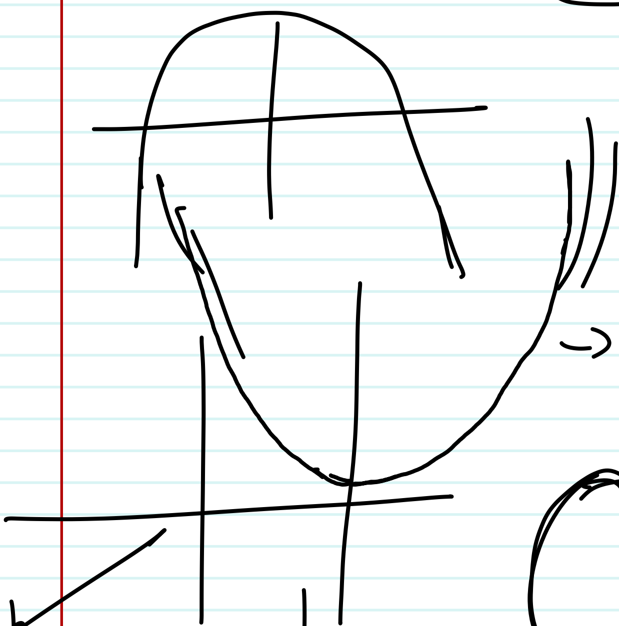
$\boxed{7200}$

4000

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4a \cdot c}{4a^2}}$$

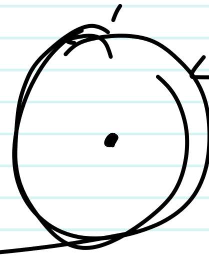
$$x + \frac{b}{2a} = \frac{\pm\sqrt{b^2 - 4ac}}{2a}$$

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$



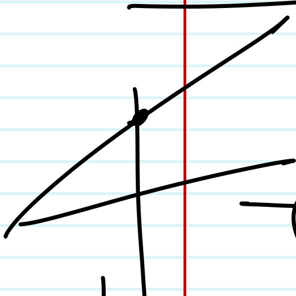
→ para

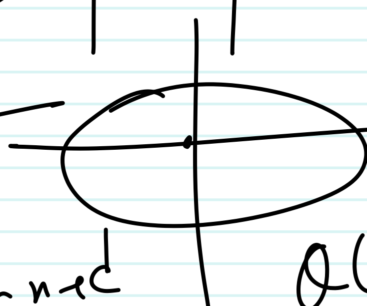
$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$



$y^2 = 4ax$

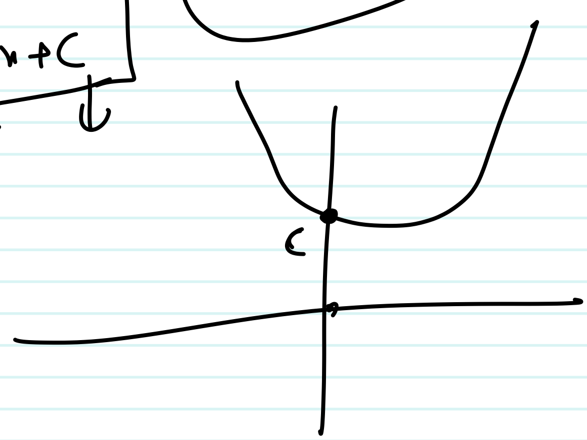
$y^2 = 2ax$





$y = mx + c$

$Q(x) = \boxed{ax^2 + bx + c}$



Ql

$Q(x) = ax^2 + bx + c$

$-\frac{b}{2a}$

$2ax + b = \tan\theta = 1 \quad \tan\theta = 1$

$\boxed{b = 1}$

$\tan(180-\theta) = -\tan\theta = 1$

ax^2+bx+c

$6x^2+4x-2=0$

$\boxed{x^2-4x-12=0}$ $\quad (-2), 6$

I

k, k+2

x+2 $\quad$ 3 $\quad$ 2n, 2n+2 $\quad$ 2n+4

x = 1

2K → K

even

K $\quad$ 2 $\quad$ 2K−1 $\quad$ 2K+1

$4k^2+4k=168$

$k^2+k=42$

$k^2+k-42=0$ $\quad 42=7\times 6$

$k^2+7k-6k-42=0$

$k(k+7)-6(k+7)=0$

$(k+7)(k-6)=0$

$k=-7,\ k=6$

Natural

$k=6,$ even $-$ $2k=2\times 6=12$

$2k+2=12+2=14$

$ax^2+bx+c=0$

$$x_1=\frac{-b+\sqrt{b^2-4ac}}{2a}$$

$$x_2=\frac{-b-\sqrt{b^2-4ac}}{2a}$$

$$x_1+x_2=\frac{-2b}{2a}=-b/a$$

$(a-b)(a+b)$

$$x_1\cdot x_2=\frac{(-b)^2-(\sqrt{b^2-4ac})^2}{(4a)^2}$$

$$=\frac{\cancel{b^2}-\cancel{b^2}+4ac}{4a^2}=c/a$$

$10x^2$

$x=1$

$a=-3$

$-a=3>0$

$a=3$

$-a=-3$

$(-a)x^2+\ldots$

$(-a)>0$

$\Rightarrow a<0$

$-2x^2+$

$10x^2,\ 20x=10$

$2x=2$

$x=1$